

Compared to What?

Team Members: _____

On March 22, 2022, four teams of students walked Pennypack Creek and collected 39 water samples, headwaters to the Delaware. One of them is hiding a road salt source. Today you will fail to find it, then find it.

Open `Class16_Compared_To_What_Skeleton.ipynb`. Answer here, on paper.

Standardizing a value means subtracting the mean and dividing by the standard deviation:

$$z = \frac{x - \bar{x}}{\sigma}$$

The result is a *z-score* — how many standard deviations a value sits from the mean. It has no units, which is the whole point.

0.1 Two students measure the same water sample. One reports chloride in mg/L, the other in parts per million. Do their z-scores differ? YES NO

In one sentence, why?

Part 1. The Wrong Outlier

20 minutes

You have been taught a rule: values more than 2 standard deviations from the mean deserve a second look; more than 3 and they are outliers. Chloride in an urban stream comes mostly from road salt, so an unusual chloride value should point you at a source. Apply the rule.

1.1 From the notebook: mean chloride _____ mg/L standard deviation _____ mg/L

Sites the rule flags, and how far downstream each one is:

_____ at _____ km _____ at _____ km

1.2 Now the three *highest* chloride samples. The largest is _____ mg/L at site _____, with $z =$ _____

Did the rule flag it? YES NO

1.3 *Commit before you plot.* Which single sample would you send someone back out to investigate as a road salt source, and which sites do you think the rule should have flagged? Write it down now; you do not get to revise it.

1.4 Run the scatter plot of chloride against distance downstream. Describe the shape of the data in one sentence — specifically, what happens around 7 km.

1.5 The rule flagged the two most pristine sites in the creek and missed the highest chloride value in the dataset. Using your plot, explain what assumption the rule makes that this data violates.

The fix is not a stricter threshold. It is a better *comparison group*. The creek changes character at the wastewater treatment plant near km 8, so compute z-scores separately above and below it — each sample judged against points that resemble it.

1.6 Upstream ($n = \rule{1cm}{0.4pt}$): flagged site $\rule{1cm}{0.4pt}$ with $z = \rule{1cm}{0.4pt}$

Does it match your answer to 1.3? YES NO

1.7 The headwater sites are no longer flagged, and nothing about their chloride concentrations changed. What changed?

1.8 Go back to the map in your mini-project notebook, or open satellite imagery for the flagged site. It sits *upstream* of the treatment plant outfall, which rules out the obvious suspect. What do you see there instead?

Part 2. Many Means

17 minutes

Different question. Your 39 samples came from one morning. If the class walked the creek again next week, how much would the *average* move?

Dr. Toran left a nitrate logger below the treatment plant recording every hour for two weeks — 330 readings. Treat those as a population and sample from it.

2.1 Look at the histogram of all 330 readings with a normal curve drawn on top. Are these data normally distributed? YES NO

Describe the shape in one sentence.

2.2 Now draw 2,000 samples of size n and histogram their means. Name *two* things that change as n goes from 1 to 6 to 24.

2.3 The narrowing follows an exact rule, $SD(\text{sample mean}) = \sigma/\sqrt{n}$. Fill in from the notebook:

n	simulated SD	$\sigma/\sqrt{n}$
6		
24		
100		

To cut the spread of the sample mean in half, n must increase by a factor of _____ (use the formula, not a guess)

2.4 The population you just sampled is not remotely bell-shaped, yet the distribution of its sample means is. State that finding as a general rule in one sentence. It has a name — use it.

Part 3. How Sure Is the Average?

8 minutes

Back to the creek. Thirty chloride samples sit downstream of the treatment plant.

3.1 From the notebook, for those 30 samples:

mean _____ mg/L standard deviation _____ mg/L standard error
_____ mg/L

95% confidence interval: _____ to _____ mg/L

3.2 Two of those numbers are about 28 and about 5. They describe completely different things. Which is which, and what does each one tell you?

3.3 The EPA drinking water standard for chloride is 250 mg/L. For each question below, circle what you would actually need in order to answer it.

“Does the *average* chloride downstream exceed the standard?”

the standard deviation

the standard error

the individual measurements

“Does *any individual location* exceed the standard?”

the standard deviation

the standard error

the individual measurements

One site in the creek does exceed 250 mg/L. Which one, and is it downstream?

Discussion

5 minutes

In Class 15 you built a null distribution by shuffling labels. Watch the same thing happen with the Pennypack nitrate data: 20,000 shuffles, testing whether nitrate differs above and below the treatment plant.

D1. Number of shuffles at least as extreme as the observed difference: _____ so $p =$

A p-value of exactly zero cannot be right — the observed result obviously *can* happen under the null, since it happened. Explain what the simulation is actually telling you, and why running a million shuffles would not fix it.

D2. That null distribution is bell-shaped, and Part 2 explains why. Standardizing the observed difference against it gives $p \approx 5 \times 10^{-7}$. In one sentence, say what standardizing let you do that counting could not — and note that this is the same move you made back in Part 1.